

Friedman's translation

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Definition 1 (First-order natural deduction). We assume given denumerably infinite sets of variables $(x, y, \dots)$, function symbols $(f, g, \dots)$, and predicate symbols $(p, q, \dots)$ with their respective arities.

First-order terms are given by:

$$t ::= f(t_1, \dots, t_n)$$

First-order formulae is given by:

$$A ::= p(t_1, \dots, t_n) \mid \perp \mid A \rightarrow A \mid A \wedge A \mid A \vee A \mid \forall x. A \mid \exists x. A$$

A *context* $(\Gamma, \Delta, \dots)$ is a finite set of formulae.

We write $\text{fv}(A)$ for the set of free variables of a formula A , defined as expected. This notion is also extended for contexts, declaring $\text{fv}(A_1, \dots, A_n) \stackrel{\text{def}}{=} \cup_{i=1}^n \text{fv}(A_i)$. We write $A\{x := t\}$ for the formula that results from substituting all the free occurrences of x in A by t .

A *sequent* is a pair of the form $\Gamma \vdash A$.

Deduction rules are:

$$\begin{array}{c} \frac{}{\Gamma, A \vdash A} \text{ax} \quad \frac{}{\Gamma \vdash A \vee \neg A} \text{lem} \quad \frac{\Gamma \vdash \perp}{\Gamma \vdash A} \text{E}\perp \quad \frac{\Gamma, A \vdash B}{\Gamma \vdash A \rightarrow B} \text{I}\rightarrow \quad \frac{\Gamma \vdash A \rightarrow B \quad \Gamma \vdash A}{\Gamma \vdash B} \text{E}\rightarrow \\[10pt] \frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \text{I}\wedge \quad \frac{\Gamma \vdash A_1 \wedge A_2}{\Gamma \vdash A_i} \text{E}\wedge_i \quad \frac{\Gamma \vdash A_i}{\Gamma \vdash A_1 \vee A_2} \text{I}\vee_i \quad \frac{\Gamma \vdash A \vee B \quad \Gamma, A \vdash C \quad \Gamma, B \vdash C}{\Gamma \vdash C} \text{E}\vee \\[10pt] \frac{\Gamma \vdash A \quad x \notin \text{fv}(\Gamma)}{\Gamma \vdash \forall x. A} \text{I}\forall \quad \frac{\Gamma \vdash \forall x. A}{\Gamma \vdash A\{x := t\}} \text{E}\forall \quad \frac{\Gamma \vdash A\{x := t\}}{\Gamma \vdash \exists x. A} \text{I}\exists \quad \frac{\Gamma \vdash \exists x. A \quad \Gamma, A \vdash B \quad x \notin \text{fv}(\Gamma, B)}{\Gamma \vdash B} \text{E}\exists \end{array}$$

As usual, we write $\neg A \stackrel{\text{def}}{=} A \rightarrow \perp$ and $(A \leftrightarrow B) \stackrel{\text{def}}{=} (A \rightarrow B) \wedge (B \rightarrow A)$.

We write $\Gamma \vdash_{\text{NK}} A$ if $\Gamma \vdash A$ is *classically valid*, i.e. derivable in the system above. We write $\Gamma \vdash_{\text{NJ}} A$ if $\Gamma \vdash A$ is *intuitionistically valid* i.e. derivable without using the **lem** rule.

Lemma 2. The following rules are admissible in NJ and NK:

$$\frac{\Gamma, A \vdash B \quad \Gamma \vdash A}{\Gamma \vdash B} \text{sub} \quad \frac{\Gamma \vdash A}{\Gamma, B \vdash A} \text{w}$$

Proof. Routine by induction on the derivation of the first premise, in each case. □

Definition 3 (Relativized negative translation). Let R be an arbitrary formula and let us write $\neg_R A \stackrel{\text{def}}{=} A \rightarrow R$. The *relativized negative translation* can be defined as follows:

$$\begin{array}{ll} p(t_1, \dots, t_n)^\bullet & \stackrel{\text{def}}{=} \neg_R \neg_R p(t_1, \dots, t_n) \\ \perp^\bullet & \stackrel{\text{def}}{=} R \\ (A \rightarrow B)^\bullet & \stackrel{\text{def}}{=} A^\bullet \rightarrow B^\bullet \\ (A \wedge B)^\bullet & \stackrel{\text{def}}{=} A^\bullet \wedge B^\bullet \\ (A \vee B)^\bullet & \stackrel{\text{def}}{=} \neg_R (\neg_R A^\bullet \wedge \neg_R B^\bullet) \\ (\forall x. A)^\bullet & \stackrel{\text{def}}{=} \forall x. A^\bullet \\ (\exists x. A)^\bullet & \stackrel{\text{def}}{=} \neg_R \forall x. \neg_R A^\bullet \end{array}$$

The translation is extended to operate on contexts by declaring that $(A_1, \dots, A_n)^\bullet \stackrel{\text{def}}{=} A_1^\bullet, \dots, A_n^\bullet$.

Remark 4. $\text{fv}(A) = \text{fv}(A^\bullet)$

Remark 5. $A\{x := t\}^\bullet = A^\bullet\{x := t\}$

Lemma 6 (Relativized explosion principle). $\Gamma, R \vdash_{\text{NJ}} A^\bullet$

Proof. By induction on A .

1. $A = p(\vec{t})$:

$$\frac{\frac{}{\Gamma, R, \neg_R p(\vec{t}) \vdash R} \text{ax}}{\Gamma, R \vdash \neg_R \neg_R p(\vec{t})} \text{I} \rightarrow$$

2. $A = \perp$:

$$\frac{}{\Gamma, R \vdash R} \text{ax}$$

3. $A = B \rightarrow C$:

$$\frac{\frac{\text{IH}}{\Gamma, R, B^\bullet \vdash C^\bullet}}{\Gamma, R \vdash B^\bullet \rightarrow C^\bullet} \text{I} \rightarrow$$

4. $A = B \wedge C$:

$$\frac{\frac{\text{IH}}{\Gamma, R \vdash B^\bullet} \quad \frac{\text{IH}}{\Gamma, R \vdash C^\bullet}}{\Gamma, R \vdash B^\bullet \wedge C^\bullet} \text{I} \wedge$$

5. $A = B \vee C$:

$$\frac{\frac{}{\Gamma, R, \neg_R B^\bullet \wedge \neg_R C^\bullet \vdash R} \text{ax}}{\Gamma, R \vdash \neg_R (\neg_R B^\bullet \wedge \neg_R C^\bullet)} \text{I} \rightarrow$$

6. $A = \forall x. B$:

$$\frac{\frac{\text{IH}}{\Gamma, R \vdash B^\bullet} \quad x \notin \text{fv}(\Gamma, R)}{\Gamma, R \vdash \forall x. B^\bullet} \text{I} \forall$$

7. $A = \exists x. B$:

$$\frac{\frac{}{\Gamma, R, \forall x. \neg_R B^\bullet \vdash R} \text{ax}}{\Gamma, R \vdash \neg_R \forall x. \neg_R B^\bullet} \text{I} \rightarrow$$

□

For technical reasons, we need to define an auxiliary operation.

Definition 7.

$$\begin{aligned} p(t_1, \dots, t_n)^\circ &\stackrel{\text{def}}{=} \neg_R p(t_1, \dots, t_n) \\ \perp^\circ &\stackrel{\text{def}}{=} \neg_R R \\ (A \rightarrow B)^\circ &\stackrel{\text{def}}{=} \neg_R (A^\bullet \rightarrow B^\bullet) \\ (A \wedge B)^\circ &\stackrel{\text{def}}{=} \neg_R (A^\bullet \wedge B^\bullet) \\ (A \vee B)^\circ &\stackrel{\text{def}}{=} \neg_R A^\bullet \wedge \neg_R B^\bullet \\ (\forall x. A)^\circ &\stackrel{\text{def}}{=} \neg_R \forall x. A^\bullet \\ (\exists x. A)^\circ &\stackrel{\text{def}}{=} \forall x. \neg_R A^\bullet \end{aligned}$$

Lemma 8. $\vdash_{\text{NJ}} A^\bullet \leftrightarrow \neg_R A^\circ$

Proof. We remark that, in general, $\vdash_{\text{NJ}} X \rightarrow \neg_R \neg_R X$. Indeed:

$$\frac{\frac{\frac{}{X, \neg_R X \vdash \neg_R X} \text{ax} \quad \frac{}{X, \neg_R X \vdash X} \text{ax}}{\neg_R X, \neg_R X \vdash R} \text{E} \rightarrow}{X \vdash \neg_R \neg_R X} \text{I} \rightarrow$$

$$\frac{}{\vdash X \rightarrow \neg_R \neg_R X} \text{I} \rightarrow$$

The proof of the lemma proceeds by induction on A :

1. $A = p(\vec{t})$: This case amounts to showing that $\vdash_{\text{NJ}} \neg_R \neg_R p(\vec{t}) \leftrightarrow \neg_R \neg_R p(\vec{t})$, which is trivial.
2. $A = \perp$: The $(\rightarrow)$ direction is an instance of the remark. For the $(\leftarrow)$ direction:

$$\begin{array}{c}
\frac{\frac{\frac{}{\neg_R \neg_R R \vdash \neg_R \neg_R R} \text{ax}}{\neg_R \neg_R R \vdash \neg_R \neg_R R} \text{ax}}{\neg_R \neg_R R \vdash R} \text{E} \rightarrow \\
\frac{}{\vdash \neg_R \neg_R R \rightarrow R} \text{I} \rightarrow
\end{array}$$

3. $A = B \rightarrow C$: The $(\rightarrow)$ direction is an instance of the remark. For the $(\leftarrow)$ direction, let $\Gamma = \neg_R \neg_R (B^\bullet \rightarrow C^\bullet), B^\bullet, C^\circ$. Then:

$$\begin{array}{c}
\frac{\frac{\frac{}{\Gamma, B^\bullet \rightarrow C^\bullet \vdash B^\bullet \rightarrow C^\bullet} \text{ax}}{\Gamma, B^\bullet \rightarrow C^\bullet \vdash C^\bullet} \text{E} \rightarrow}{\Gamma, B^\bullet \rightarrow C^\bullet \vdash \neg_R C^\circ} \text{IH} \frac{}{\Gamma, B^\bullet \rightarrow C^\bullet \vdash C^\circ} \text{ax} \\
\frac{}{\Gamma, B^\bullet \rightarrow C^\bullet \vdash R} \text{E} \rightarrow \\
\frac{\frac{}{\Gamma \vdash \neg_R \neg_R (B^\bullet \rightarrow C^\bullet)} \text{ax}}{\Gamma \vdash \neg_R (B^\bullet \rightarrow C^\bullet)} \text{I} \rightarrow \\
\frac{}{\neg_R \neg_R (B^\bullet \rightarrow C^\bullet), B^\bullet, C^\circ \vdash R} \text{E} \rightarrow \\
\frac{}{\neg_R \neg_R (B^\bullet \rightarrow C^\bullet), B^\bullet \vdash \neg_R C^\circ} \text{I} \rightarrow \\
\frac{}{\neg_R \neg_R (B^\bullet \rightarrow C^\bullet), B^\bullet \vdash C^\bullet} \text{IH} \\
\frac{}{\neg_R \neg_R (B^\bullet \rightarrow C^\bullet) \vdash B^\bullet \rightarrow C^\bullet} \text{I} \rightarrow \\
\frac{}{\vdash \neg_R \neg_R (B^\bullet \rightarrow C^\bullet) \rightarrow (B^\bullet \rightarrow C^\bullet)} \text{I} \rightarrow
\end{array}$$

4. $A = B \wedge C$: The $(\rightarrow)$ direction is an instance of the remark. For the $(\leftarrow)$ direction, let $\Gamma_1 = \neg_R \neg_R (B^\bullet \wedge C^\bullet), B^\circ, B^\bullet \wedge C^\bullet$ and let π_1 be given by:

$$\begin{array}{c}
\frac{\frac{\frac{}{\Gamma \vdash B^\bullet \wedge C^\bullet} \text{ax}}{\Gamma \vdash B^\bullet} \text{E} \wedge_1}{\Gamma \vdash \neg B^\circ} \text{IH} \frac{}{\Gamma \vdash B^\circ} \text{ax} \\
\frac{}{\neg_R \neg_R (B^\bullet \wedge C^\bullet), B^\circ, B^\bullet \wedge C^\bullet \vdash R} \text{E} \rightarrow \\
\frac{\frac{}{\neg_R \neg_R (B^\bullet \wedge C^\bullet), B^\circ \vdash \neg_R \neg_R (B^\bullet \wedge C^\bullet)} \text{ax}}{\neg_R \neg_R (B^\bullet \wedge C^\bullet), B^\circ \vdash \neg_R (B^\bullet \wedge C^\bullet)} \text{I} \rightarrow \\
\frac{}{\neg_R \neg_R (B^\bullet \wedge C^\bullet), B^\circ \vdash R} \text{E} \rightarrow \\
\frac{}{\neg_R \neg_R (B^\bullet \wedge C^\bullet) \vdash \neg_R B^\circ} \text{I} \rightarrow \\
\frac{}{\neg_R \neg_R (B^\bullet \wedge C^\bullet) \vdash B^\bullet} \text{IH}
\end{array}$$

Symmetrically, let π_2 be a derivation of $\neg_R \neg_R (B^\bullet \wedge C^\bullet) \vdash C^\bullet$. Then:

$$\begin{array}{c}
\frac{\frac{}{\neg_R \neg_R (B^\bullet \wedge C^\bullet) \vdash B^\bullet} \pi_1}{\neg_R \neg_R (B^\bullet \wedge C^\bullet) \vdash B^\bullet \wedge C^\bullet} \text{I} \wedge \\
\frac{}{\vdash \neg_R \neg_R (B^\bullet \wedge C^\bullet) \rightarrow (B^\bullet \wedge C^\bullet)} \text{I} \rightarrow
\end{array}$$

5. $A = B \vee C$: This case amounts to showing that $\vdash_{\text{NJ}} \neg_R (\neg_R B^\bullet \wedge \neg_R C^\bullet) \leftrightarrow \neg_R (\neg_R B^\bullet \wedge \neg_R C^\bullet)$, which is trivial.
6. $A = \forall x. B$: The $(\rightarrow)$ direction is an instance of the remark. For the $(\leftarrow)$ direction, let $\Gamma = \neg_R \neg_R \forall x. B^\bullet, B^\circ, \forall x. B^\bullet$.

Then:

$$\begin{array}{c}
\frac{}{\Gamma \vdash \forall x. B^\bullet} \text{ax} \\
\frac{}{\Gamma \vdash B^\bullet} \text{E}\forall \\
\frac{}{\Gamma \vdash \neg_R B^\circ} \text{IH} \quad \frac{}{\Gamma \vdash B^\circ} \text{ax} \\
\frac{}{\neg_R \neg_R \forall x. B^\bullet, B^\circ \vdash \neg_R \neg_R \forall x. B^\bullet} \text{ax} \quad \frac{}{\neg_R \neg_R \forall x. B^\bullet, B^\circ, \forall x. B^\bullet \vdash R} \text{E}\rightarrow \\
\frac{}{\neg_R \neg_R \forall x. B^\bullet, B^\circ \vdash \neg_R \forall x. B^\bullet} \text{I}\rightarrow \\
\frac{}{\neg_R \neg_R \forall x. B^\bullet, B^\circ \vdash R} \text{E}\rightarrow \\
\frac{}{\neg_R \neg_R \forall x. B^\bullet \vdash \neg_R B^\circ} \text{I}\rightarrow \\
\frac{}{\neg_R \neg_R \forall x. B^\bullet \vdash B^\bullet} \text{IH} \quad x \notin \text{fv}(\neg_R \neg_R \forall x. B^\bullet) \\
\frac{}{\neg_R \neg_R \forall x. B^\bullet \vdash \forall x. B^\bullet} \text{I}\forall \\
\frac{}{\vdash \neg_R \neg_R \forall x. B^\bullet \rightarrow \forall x. B^\bullet} \text{I}\rightarrow
\end{array}$$

7. $A = \exists x. B$: This case amounts to showing that $\vdash_{\text{NJ}} \neg_R \forall x. \neg_R B^\bullet \leftrightarrow \neg_R \forall x. \neg_R B^\bullet$, which is trivial.

□

Lemma 9 (Soundness of the translation). If $\Gamma \vdash_{\text{NK}} A$ then $\Gamma^\bullet \vdash_{\text{NJ}} A^\bullet$.

Proof. By induction on the derivation of $\Gamma \vdash_{\text{NK}} A$.

1. **ax**: Let $\Gamma, A \vdash A$. Then:

$$\frac{}{\Gamma^\bullet, A^\bullet \vdash A^\bullet} \text{ax}$$

2. **lem**: Let $\Gamma \vdash A \vee \neg A$. Then:

$$\begin{array}{c}
\frac{}{\Gamma^\bullet, \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet \vdash \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet} \text{ax} \quad \frac{}{\Gamma^\bullet, \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet \vdash \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet} \text{ax} \\
\frac{}{\Gamma^\bullet, \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet \vdash \neg_R \neg_R A^\bullet} \text{E}\wedge_2 \quad \frac{}{\Gamma^\bullet, \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet \vdash \neg_R A^\bullet} \text{E}\wedge_1 \\
\frac{}{\Gamma^\bullet, \neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet \vdash R} \text{E}\rightarrow \\
\frac{}{\Gamma^\bullet \vdash \neg_R(\neg_R A^\bullet \wedge \neg_R \neg_R A^\bullet)} \text{I}\rightarrow
\end{array}$$

3. **E $\perp$** : Let $\Gamma \vdash A$ be derived from $\Gamma \vdash \perp$. Then:

$$\frac{\frac{}{\Gamma^\bullet, R \vdash A^\bullet} \text{Lem. 6} \quad \frac{}{\Gamma^\bullet \vdash R} \text{IH}}{\Gamma^\bullet \vdash A^\bullet} \text{sub}$$

4. **I $\rightarrow$** : Straightforward by resorting to the IH.

5. **E $\rightarrow$** : Straightforward by resorting to the IH.

6. **I $\wedge$** : Straightforward by resorting to the IH.

7. **E $\wedge_i$** : Straightforward by resorting to the IH.

8. **I $\vee_i$** : Let $\Gamma \vdash A_1 \vee A_2$ be derived from $\Gamma \vdash A_i$. Then:

$$\begin{array}{c}
\frac{}{\Gamma^\bullet, \neg_R A_1^\bullet \wedge \neg_R A_2^\bullet \vdash \neg_R A_1^\bullet \wedge \neg_R A_2^\bullet} \text{ax} \quad \frac{}{\Gamma^\bullet \vdash A_i^\bullet} \text{IH} \\
\frac{}{\Gamma^\bullet, \neg_R A_1^\bullet \wedge \neg_R A_2^\bullet \vdash \neg_R A_i^\bullet} \text{E}\wedge_i \quad \frac{}{\Gamma^\bullet, \neg_R A_1^\bullet \wedge \neg_R A_2^\bullet \vdash A_i^\bullet} \text{w} \\
\frac{}{\Gamma^\bullet, \neg_R A_1^\bullet \wedge \neg_R A_2^\bullet \vdash R} \text{E}\rightarrow \\
\frac{}{\Gamma^\bullet \vdash \neg_R(\neg_R A_1^\bullet \wedge \neg_R A_2^\bullet)} \text{I}\rightarrow
\end{array}$$

9. **E $\vee$** : Let $\Gamma \vdash C$ be derived from $\Gamma \vdash A \vee B$ and $\Gamma, A \vdash C$ and $\Gamma, B \vdash C$. Let π_1 be the following derivation:

$$\begin{array}{c}
\frac{}{\Gamma^\bullet, A^\bullet \vdash C^\bullet} \text{IH} \\
\frac{}{\Gamma^\bullet, C^\circ, A^\bullet \vdash C^\bullet} \text{w} \\
\frac{}{\Gamma^\bullet, C^\circ, A^\bullet \vdash \neg_R C^\circ} \text{Lem. 8} \quad \frac{}{\Gamma^\bullet, C^\circ, A^\bullet \vdash C^\circ} \text{ax} \\
\frac{}{\Gamma^\bullet, C^\circ, A^\bullet \vdash R} \text{E}\rightarrow \\
\frac{}{\Gamma^\bullet, C^\circ \vdash \neg_R A^\bullet} \text{I}\rightarrow
\end{array}$$

Symmetrically, let π_2 be a derivation of $\Gamma^\bullet, C^\circ \vdash \neg_R B^\bullet$. Then:

$$\begin{array}{c}
\frac{\text{IH}}{\Gamma^\bullet \vdash \neg_R(\neg_R A^\bullet \wedge \neg_R B^\bullet)} \quad \frac{\pi_1}{\Gamma^\bullet, C^\circ \vdash \neg_R A^\bullet} \quad \frac{\pi_2}{\Gamma^\bullet, C^\circ \vdash \neg_R B^\bullet} \\
\frac{\Gamma^\bullet, C^\circ \vdash \neg_R(\neg_R A^\bullet \wedge \neg_R B^\bullet)}{\Gamma^\bullet, C^\circ \vdash \neg_R A^\bullet \wedge \neg_R B^\bullet} \text{w} \quad \frac{\Gamma^\bullet, C^\circ \vdash \neg_R A^\bullet \wedge \neg_R B^\bullet}{\Gamma^\bullet, C^\circ \vdash R} \text{I}\wedge \\
\frac{\Gamma^\bullet, C^\circ \vdash R}{\Gamma^\bullet \vdash \neg_R C^\circ} \text{E}\rightarrow \\
\frac{\Gamma^\bullet \vdash \neg_R C^\circ}{\Gamma^\bullet \vdash C^\bullet} \text{I}\rightarrow \\
\hline
\Gamma^\bullet \vdash C^\bullet \quad \text{Lem. 8}
\end{array}$$

10. $\text{I}\forall$: Straightforward by resorting to the IH.

11. $\text{E}\forall$: Straightforward by resorting to the IH and Rem. 5.

12. $\text{I}\exists$: Let $\Gamma \vdash \exists x. A$ be derived from $\Gamma \vdash A\{x := t\}$. Recall from Rem. 5 that $A\{x := t\}^\bullet = A^\bullet\{x := t\}$. Then:

$$\begin{array}{c}
\frac{\text{IH}}{\Gamma^\bullet, \forall x. \neg_R A^\bullet \vdash \forall x. \neg_R A^\bullet} \text{ax} \quad \frac{\Gamma^\bullet \vdash A^\bullet\{x := t\}}{\Gamma^\bullet, \forall x. \neg_R A^\bullet \vdash A^\bullet\{x := t\}} \text{w} \\
\frac{\Gamma^\bullet, \forall x. \neg_R A^\bullet \vdash \neg_R A^\bullet\{x := t\}}{\Gamma^\bullet, \forall x. \neg_R A^\bullet \vdash R} \text{E}\forall \quad \frac{\Gamma^\bullet, \forall x. \neg_R A^\bullet \vdash R}{\Gamma^\bullet \vdash \neg_R \forall x. \neg_R A^\bullet} \text{E}\rightarrow \\
\hline
\Gamma^\bullet \vdash \neg_R \forall x. \neg_R A^\bullet \quad \text{I}\rightarrow
\end{array}$$

13. $\text{E}\exists$: Let $\Gamma \vdash B$ be derived from $\Gamma \vdash \exists x. A$ and $\Gamma, A \vdash B$, where $x \notin \text{fv}(\Gamma, B)$. Then:

$$\begin{array}{c}
\text{IH} \\
\frac{\Gamma^\bullet, A^\bullet \vdash B^\bullet}{\Gamma^\bullet, B^\circ, A^\bullet \vdash B^\bullet} \text{w} \\
\frac{\Gamma^\bullet, B^\circ, A^\bullet \vdash B^\bullet}{\Gamma^\bullet, B^\circ, A^\bullet \vdash \neg_R B^\circ} \text{Lem. 8} \quad \frac{\Gamma^\bullet, B^\circ, A^\bullet \vdash B^\circ}{\Gamma^\bullet, B^\circ, A^\bullet \vdash R} \text{ax} \\
\frac{\Gamma^\bullet, B^\circ, A^\bullet \vdash R}{\Gamma^\bullet, B^\circ \vdash \neg_R A^\bullet} \text{E}\rightarrow \\
\text{IH} \quad \frac{\Gamma^\bullet \vdash \neg_R \forall x. \neg_R A^\bullet}{\Gamma^\bullet, B^\circ \vdash \neg_R \forall x. \neg_R A^\bullet} \text{w} \quad \frac{\Gamma^\bullet, B^\circ \vdash \neg_R A^\bullet}{\Gamma^\bullet, B^\circ \vdash \forall x. \neg_R A^\bullet} \text{I}\rightarrow \quad \frac{\Gamma^\bullet, B^\circ \vdash \forall x. \neg_R A^\bullet}{\Gamma^\bullet, B^\circ \vdash R} \text{I}\forall \\
\frac{\Gamma^\bullet, B^\circ \vdash R}{\Gamma^\bullet \vdash \neg_R B^\circ} \text{E}\rightarrow \\
\frac{\Gamma^\bullet \vdash \neg_R B^\circ}{\Gamma^\bullet \vdash B^\bullet} \text{I}\rightarrow \\
\hline
\Gamma^\bullet \vdash B^\bullet \quad \text{Lem. 8}
\end{array}$$

□

Theorem 10. If $\vdash_{\text{NK}} \exists x. p(\vec{t})$ then $\vdash_{\text{NJ}} \exists x. p(\vec{t})$.

Proof. Let $\vdash_{\text{NK}} \exists x. p(\vec{t})$. By Lem. 9 we have that $\vdash_{\text{NJ}} \neg_R \forall x. \neg_R \neg_R \neg_R p(\vec{t})$, This is equivalent to $\vdash_{\text{NJ}} \neg_R \forall x. \neg_R p(\vec{t})$, since, in general, $\vdash_{\text{NJ}} \neg_R X \leftrightarrow \neg_R \neg_R \neg_R X$. The reasoning so far holds for an arbitrary R . In particular, taking $R := \exists x. p(\vec{t})$ we have that $\vdash_{\text{NJ}} \forall x. (p(\vec{t}) \rightarrow \exists x. p(\vec{t})) \rightarrow \exists x. p(\vec{t})$. It is easy to show that the antecedent holds, as it essentially corresponds to the introduction rule of the existential, which means that indeed $\vdash_{\text{NJ}} \exists x. p(\vec{t})$. □